AMA572 · Assignment 2

Complete Solutions · Due: October 16, 2026

Q1(a). First-passage transform and total firm value

Use the risk-adjusted measure of Lecture 4. Let $r, m, \sigma, P > 0$, $0 < V_B < V$, $0 \le \alpha \le 1$, $\rho \ge 0$, and $0 \le \kappa < 1$. Here P is total debt face value, ρ the coupon rate, κ the tax rate, and δ the asset payout rate. The model is

$$\frac{dV_t}{V_t} = (r-\delta)\,dt + \sigma\,dW_t, \qquad \tau = \inf\{t \ge 0 : V_t \le V_B\}, \quad V_0 = V.$$

Define, for $q > 0$,

$$\xi = \frac{r-\delta}{\sigma^2} - \frac{1}{2}, \qquad \eta_q = \sqrt{\xi^2 + \frac{2q}{\sigma^2}} + \xi, \qquad L_q = \mathbb{E}[e^{-q\tau}].$$

We set $e^{-q\tau} = 0$ on $\{\tau = \infty\}$; eventual default need not have probability one. Since the asset paths are continuous, $V_\tau = V_B$ whenever $\tau < \infty$.

First-passage transform. The logarithmic asset process and the lower barrier are

$$\log \frac{V_t}{V} = \mu t + \sigma W_t, \qquad \mu = r - \delta - \tfrac{1}{2}\sigma^2 = \sigma^2 \xi, \qquad \ell = \log \frac{V_B}{V} < 0.$$

Applying the $b < 0$ case of Appendix (A.1), with $b = \ell$, gives

$$\begin{aligned} L_q &= \exp\left\{ \frac{\ell}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 q} + \mu\right)\right\} \\ &= \exp\{\ell \eta_q\} = \left(\frac{V_B}{V}\right)^{\eta_q}. \end{aligned}$$

Total firm value. Add the present value of tax benefits to V and subtract bankruptcy costs. Using $\int_0^\tau e^{-rt}dt = (1 - e^{-r\tau})/r$,

$$\begin{aligned} v(V; V_B) &= V + \kappa\rho P\, \mathbb{E}\left[\int_0^\tau e^{-rt}dt\right] - (1-\alpha) V_B L_r \\ &= V + \frac{\kappa\rho P}{r}(1 - L_r) - (1-\alpha) V_B L_r. \end{aligned}$$

Substituting the first-passage transform proves Lecture 4, equation (4):

$$\boxed{v(V; V_B) = V + \frac{\kappa\rho}{r} P - \left[\frac{\kappa\rho}{r} P + (1-\alpha) V_B\right]\left(\frac{V_B}{V}\right)^{\eta_r}.}$$

Here $\eta_r = \sqrt{\xi^2 + 2r/\sigma^2} + \xi$. No numerical parameter values are specified in the assignment, so the required answer is symbolic.

Q1(b). Total debt value

A unit-face-value bond with maturity T receives principal if it survives, recovery if default occurs before maturity, and coupons until $\tau \wedge T$. The discounted recovery is $e^{-r\tau}ce^{-r(T-\tau)} = ce^{-rT}$. Thus

$$B(V; V_B, T) = \mathbb{E}\left[e^{-rT}\mathbf{1}_{\{T<\tau\}} + ce^{-rT}\mathbf{1}_{\{\tau\le T\}} + \rho\int_0^{\tau\wedge T} e^{-rt}dt\right],$$
$$D(V; V_B) = P\int_0^\infty me^{-mT}B(V; V_B, T)\,dT, \qquad c = \frac{r+m}{m}\frac{\alpha V_B}{P}.$$

Integrate the three nonnegative cash-flow terms separately. Tonelli's theorem permits exchanging integrals and expectations. All three expressions below are **per unit of total face value**.

Coupons. For a fixed payment time t, only maturities $T \ge t$ contribute:

$$\begin{aligned} I_C &= \rho\,\mathbb{E}\left[\int_0^\tau e^{-rt}\left(\int_t^\infty me^{-mT}dT\right)dt\right] \\ &= \rho\,\mathbb{E}\left[\int_0^\tau e^{-(r+m)t}dt\right] = \frac{\rho}{r+m}(1-L_{r+m}). \end{aligned}$$

Principal at maturity and **recovery after default** give, respectively,

$$I_M = \mathbb{E}\left[\int_0^\tau me^{-(r+m)T}dT\right] = \frac{m}{r+m}(1-L_{r+m}),$$
$$I_R = c\,\mathbb{E}\left[\int_\tau^\infty me^{-(r+m)T}dT\right] = \frac{cm}{r+m}L_{r+m}.$$

Using $D = P(I_C + I_M + I_R)$ and $Pcm/(r+m) = \alpha V_B$ yields

$$\boxed{D(V; V_B) = P\frac{\rho+m}{r+m} - \left[P\frac{\rho+m}{r+m} - \alpha V_B\right]\left(\frac{V_B}{V}\right)^{\eta_{r+m}}.}$$

To recover exactly the coefficient in Lecture 4, equation (5), observe that

$$\frac{\rho+m}{r+m} - \frac{\alpha V_B}{P} = \frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{r+m}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}.$$

Consequently,

$$\boxed{\begin{aligned} D(V; V_B) = {} & P\frac{\rho+m}{r+m} \\ & - P\left[\frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{r+m}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right]\left(\frac{V_B}{V}\right)^{\eta_{r+m}}. \end{aligned}}$$

This proves (5), since $\eta_{r+m} = \sqrt{\xi^2 + 2(r+m)/\sigma^2} + \xi$. As a boundary check, $v(V_B; V_B) = D(V_B; V_B) = \alpha V_B$, hence equity is zero at default.

Q2. Smooth-pasting condition and the default barrier

Introduce the abbreviations

$$a = \eta_r, \qquad b = \eta_{r+m}, \qquad K = \frac{\kappa\rho}{r}, \qquad H = \frac{\rho+m}{r+m}.$$

The coefficient identity in Q1(b) reduces the given equity value to

$$S(V;V_B) = V + KP - [KP + (1-\alpha)V_B]\left(\frac{V_B}{V}\right)^a - PH + [PH - \alpha V_B]\left(\frac{V_B}{V}\right)^b.$$

Differentiate while holding V_B and P fixed. For a constant exponent q,

$$\frac{\partial}{\partial V}\left(\frac{V_B}{V}\right)^q = -\frac{q}{V}\left(\frac{V_B}{V}\right)^q.$$

It follows that, in the continuation region $V > V_B$,

$$S_V(V;V_B) = 1 + \frac{a[KP + (1-\alpha)V_B]}{V}\left(\frac{V_B}{V}\right)^a - \frac{b[PH - \alpha V_B]}{V}\left(\frac{V_B}{V}\right)^b.$$

Now take the limit $V \downarrow V_B$, so both power terms equal one:

$$\begin{aligned} S_V(V_B+;V_B) &= 1 + \frac{a[KP + (1-\alpha)V_B]}{V_B} - \frac{b[PH - \alpha V_B]}{V_B} \\ &= 1 + (1-\alpha)a + \alpha b + \frac{P}{V_B}(Ka - Hb). \end{aligned}$$

Setting this expression to zero and multiplying by V_B gives

$$[1 + (1-\alpha)a + \alpha b]\, V_B = P(Hb - Ka).$$

Since $a, b > 0$ and $0 \le \alpha \le 1$, the denominator is positive. The resulting solution is

$$\boxed{V_B^* = \epsilon P, \qquad \epsilon = \frac{Hb - Ka}{1 + (1-\alpha)a + \alpha b}.}$$

Substituting the definitions of a, b, K, H gives precisely the expression requested:

$$\epsilon = \frac{\frac{\rho+m}{r+m}\left(\sqrt{\xi^2 + \frac{2(r+m)}{\sigma^2}} + \xi\right) - \frac{\kappa\rho}{r}\left(\sqrt{\xi^2 + \frac{2r}{\sigma^2}} + \xi\right)}{1 + (1-\alpha)\left(\sqrt{\xi^2 + \frac{2r}{\sigma^2}} + \xi\right) + \alpha\left(\sqrt{\xi^2 + \frac{2(r+m)}{\sigma^2}} + \xi\right)}.$$

The order of operations matters: first take the partial derivative in V, then evaluate at the barrier. Substituting $V = V_B$ first gives the identity $S(V_B;V_B) = 0$ and does not establish smooth pasting. The result above proves the equation asked in Q2; the constrained global-optimality claim is the separate Theorem 1(c) of Lecture 4.

References: Assignment 2, pp. 1-2 and Appendix (A.1); Lecture 4, Proposition 1, pp. 3-5, and Theorem 1(a), pp. 6-8. The lecture's economic restrictions $0 \le c \le 1$ and $\kappa\rho < r$ are understood.